Seat No.

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Midterm Examination of Semester 1 — **Year**: 2025

Subject: 031001133 Mathematics III — **Section**: 15-17

Date: Wednesday, August 6, 2025 — **Time**: 08.00 – 10.00

Examination Setter Name: Assoc. Prof. Dr. Suphawat Asawasamrit
Dr. Chaiyod Kamthorncharoen

Student Name ______________________ ID ____________ Class ______ No ___

<u>Directions;</u>

1. This examination paper has not been edited. If any errors are identified, students are advised to exercise discretion in addressing them.
2. The examination consists of 8 questions on 11 pages (including the cover page). The total score is 77 points.
3. Complete all questions on this examination paper.
4. Books, documents, and lecture notes are not allowed.
5. Electronic calculators are not allowed.
6. Students must remain in the examination room for no less than 1 hour from the start of the examination.
7. Do not open or start the examination before permission is granted. Students must strictly follow the instructions of the examination.
8. Access to restroom facilities during the examination is not permitted, except in cases of emergency.
9. Students are not allowed to take the examination paper or copy the examination paper out of the examination room. Otherwise, it will be considered cheating.
10. Usage of all communication devices during the examination is strictly prohibited.
11. Smart watches (electronic watches) are not permitted during the exam.

> Cheating in examinations is considered a serious offense and carries the penalty of expulsion from student status.

No.	1.	2.	3.	4.	5.	6.	7.	8.	Total
score									

This examination is part of the assessment of students' abilities as follows:		Q1- Q8
CLO 1	Demonstrate comprehensive understanding of complex numbers, matrices, sequences, series, and vectors, including their meanings and underlying principles.	√
CLO 2	Apply the acquired knowledge and understanding of mathematics to solve a diverse range of problems effectively and appropriately.	√
CLO 3	Proficiently calculate, analyze, and employ fundamental mathematical principles in the context of scientific disciplines..	√
CLO 4	Cultivate moral and ethical values, respect for others' rights, actively listen to diverse opinions, and develop effective teamwork skills.	√
CLO 5	Practice and enhance the computational skills through the proficient use of information technology tools.	√

1.

1.1 Simplify $\dfrac{2i^{81}}{1-i^{31}} + \dfrac{4i^{72}-5i^{19}}{i^{22}}$ (answer in $a+bi$ form). (4 points)

1.2 Write $\left(\sqrt{2}-\sqrt{2}i\right)^{-6}$ in the form of $a+bi$ where a and b are real numbers. (5 points)

2. Assume that z is the complex number such that

$$|z| - 4\bar{z} + 5z = \frac{4i^{35}\left(2i^{13} + 9i^{92}\right)}{i^{28}}.$$

Evaluate z^{-1} and $\left|\frac{z^2}{z^{-1}z}\right|$. (Hint: let $z = x + yi$) (8 points)

3. Let z_0, z_1, and z_2 be 3[rd] roots of $8i$, where $k = 0$, 1, and 2 respectively. Find $z_0 \cdot z_1 \cdot z_2$.

(10 points)

4. Find all solutions to $x^7 + 2x^6 - 64x - 128 = 0$. (10 points)

5.

5.1 Given that $\frac{3}{2}\begin{bmatrix}-2a & b \\ 3c & -d\end{bmatrix} + \frac{1}{2}\begin{bmatrix}-8 & 6 \\ 4 & 7\end{bmatrix} - 3\begin{bmatrix}3 & -1 \\ 0 & 2\end{bmatrix} = \begin{bmatrix}0 & 0 \\ 11 & 0\end{bmatrix}$.

Determine the value of $a+b+c+d$. (5 points)

5.2 Let $A = \begin{bmatrix}3 & -1 & 0 \\ -1 & 1 & 2 \\ 1 & 0 & -3\end{bmatrix}$. Find the symmetric matrix (S) of A^2. (5 points)

6. Evaluate A^{-1}, where $A \cdot \begin{bmatrix} 3 & 0 & 1 \\ 5 & -1 & 2 \\ -4 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 4 & -2 & -3 \\ 0 & 3 & 2 \\ -1 & 2 & 3 \end{bmatrix}$. (10 points)

7. Assume that A, B, C, and I_2 are 2×2 matrices where I_2 is the identity matrix.

If $\det(\sqrt{2}A) = 24$, $\det(C^{-1}) = 2$, and $AB^TC = \begin{bmatrix} 6 & -1 \\ 4 & 2 \end{bmatrix}$, find $\det(B)$. (10 points)

8.1 Solve the system of equations **using Gauss-Jordan elimination** (5 points)

$$
\begin{aligned}
x_1 + 2x_2 - x_3 &= 4 \\
-2x_1 + 3x_2 - x_3 &= 4 \\
-4x_1 + 2x_2 &= -2.
\end{aligned}
$$

8.2 Find x_3, using Cramer's rule (5 points)

$$\begin{aligned}
2x_1 \quad\quad\quad + x_3 - x_4 &= 0 \\
x_1 + x_2 - x_3 \quad\quad\quad &= 0 \\
x_1 + x_2 + x_3 + x_4 &= 11 \\
x_1 \quad\quad\quad - x_3 - x_4 &= 0.
\end{aligned}$$